

Test 16: AB-cloud GUE classification

Verdict: PASS | Generated: 2026-09-02 14:32:00 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Statistical: χ^2 over the central bulk window vs GUE 0.5992, MC p-value as diagnostics.

METHOD: Statistical: χ^2 over the central bulk window vs GUE 0.5992, MC p-value as diagnostics.

Test 16 HARDCORE pass 2: 96x96, n=5 realizations at $\rho=0.5000$, $N_v=2$, $q=1.000$, $W_{\text{eff}}=1.00$, ensemble deep checks (bootstrap CI, scatter, sign count).

Test 16 HARDCORE: $\rho = 0.5798 - 0.0037$ (n=5), bootstrap CI [0.5732, 0.5859] misses GUE, scatter $\rho = 0.0082$.

Test 16b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/plot_02' (.png 600dpi / .svg / .pdf)

- 'plots/plot_03' (.png 600dpi / .svg / .pdf)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v19.jl --test 16 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

```
[14:06:53.826] [+1129.108s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental
```

[14:11:12.341] [+1387.623s] calc: ?r? = 0.572908 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

FULL COMPUTATION LOG

FILE COMPUTATION LOG

[14:16:14.259] [+1689.541s] calc: ?r? = 0.569121 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

```
[14:16:19.224] [+1694.506s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental
```

[14:21:34.935] [+2010.217s] calc: ?r? = 0.586869 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[14:21:38.664] [+2013.946s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[14:26:41.417] [+2316.699s] calc: ?r? = 0.584535 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

```
[14:26:47.323] [+2322.605s] build ab cloud hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental
```

[14:31:55.621] [+2630.903s] calc: ?r? = 0.585615 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

??

TEST 16 (pass 2, HARDCORE): ?r ? GUE classification ? deep validation

Deep ensemble at the launch-standard config (pass 1 already swept breadth)

CONSOLE CAPTURE

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? RUNTIME: n=5 realizations at 96x96 (N=9216) ? 1.5?3 min each ? expect ? 12?17 min

Config: ?=0.5000, Nv=2, q=1.000, BC=:open, W_eff=1.00 (standard cap 1.0)

real 1/5: ?r? = 0.5729

real 2/5: ?r? = 0.5691

real 3/5: ?r? = 0.5869

real 4/5: ?r? = 0.5845

real 5/5: ?r? = 0.5856

Ensemble: ?r? = 0.5798 – 0.0037 (n=5)

Bootstrap 95% CI [0.5732, 0.5859] ? GUE (diagnostic); above/below GUE: 0/5

??

?r? > 0.55 PASS ? ?r? = 0.5798 – 0.0037 vs GUE 0.5992 (n=5)

scatter ? 0.022 PASS ? ?=0.0082 ? ? monograph band

Test 16b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS